

FACULTY OF
COMPUTER SCIENCE

Otto-von-Guericke-University Magdeburg

Faculty of Computer Science

Department of Databases and Software Engineering

Comparative Analysis of Multimodal Emotion Detection Approaches

Bachelor thesis

Author:

Lukas Eichel

Examiner:

Prof. Dr. rer. nat. habil. Gunter Saake

2nd Examiner and Supervisor:

Dr.-Ing. David Broneske

Magdeburg, 13.09.2024

Abstract

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

Contents

1. Introduction **1**
- 1.1. Motivation . . . 1
- 1.2. Structure . . . 1

2. Background **2**
- 2.1. Emotion . . . 2
  - 2.1.1. Emotion Classification . . . 3
  - 2.1.2. Emotion in Text . . . 4
  - 2.1.3. Emotion in Speech . . . 5
  - 2.1.4. Emotion in Facial Expression . . . 6
- 2.2. Classification Approaches . . . 8
  - 2.2.1. Classification in Text . . . 8
  - 2.2.2. Classification in Audio . . . 13
  - 2.2.3. Classification in Images and Videos . . . 16
- 2.3. Datasets . . . 19

3. Related Work **21**
- 3.1. Audio and Video Based . . . 21
- 3.2. Audio, Video and Text Based . . . 22

4. Implementation **25**
- 4.1. . . . 25
- 4.2. Prompt Engineering for LLM . . . 25
- 4.3. Model Architectures . . . 25
- 4.4. Training . . . 25
- 4.5. Scoring and Weighting . . . 25

Bibliography **27**

Chapter 1.

Introduction

1.1. Motivation

find motivation

1.2. Structure

structure

Chapter 2.

Background

This chapter will give an introduction and overview on different approaches to classify types of emotions with respect to their advantages and disadvantages. Furthermore it will explain the approaches and assumptions to recognize emotion in text, video and audio. Based on this common implementation techniques for supervised and unsupervised approaches are presented with a focus on convolution and transformer models since they are the leading architectures in deep learning. At the end, datasets are introduced which are the foundation for the later implementation and evaluation of the models.

2.1. Emotion

There is no unambigious definition for the term ‘emotion’. There are many different emotions, with different level of intensity. Considering the question of what an emotion is and where it comes from, William James discussed two different theories about the origin of them. Either there are seperate areas in the cerebrum or else they correspond to motory and sensory processes [1]. If they relie on the seperate areas, the research would be much easier. They could observate these regions while testing different actions and influences on the probants. The results should be quite similar between most people. However it is most likely the case, that emotion correspond to these motory and sensory processes. They will be different from human to human in intensity and execution, so researchers have to study the small similarieties and also the possible patterns and variations from the shown processes. In this case these factors and all brain activitys must be captured to set a prove for it. Many work in psychology add further aspects to the theory. In order to provide a framework, the following chapters will focus just on the given emotion and how to recognize it in written text, speech and facial expressions. These are the most common expressive features captured by emotion theorists, but there are many more like gestures or going into a medical direction the blood preassure and other factors might have an approach for determining the leading emotion.

In the other direction there are first approaches to simulate sentiments with computer or robots like animatronics. Learned patterns can be applied from a simple command to emotion features which are currently still very overacted.

Figure 2.1 presents that information like motory, sensory processes, their combination and the differentiation from semantic and syntatic context are categorized in high-level and low level information to process an emotion.

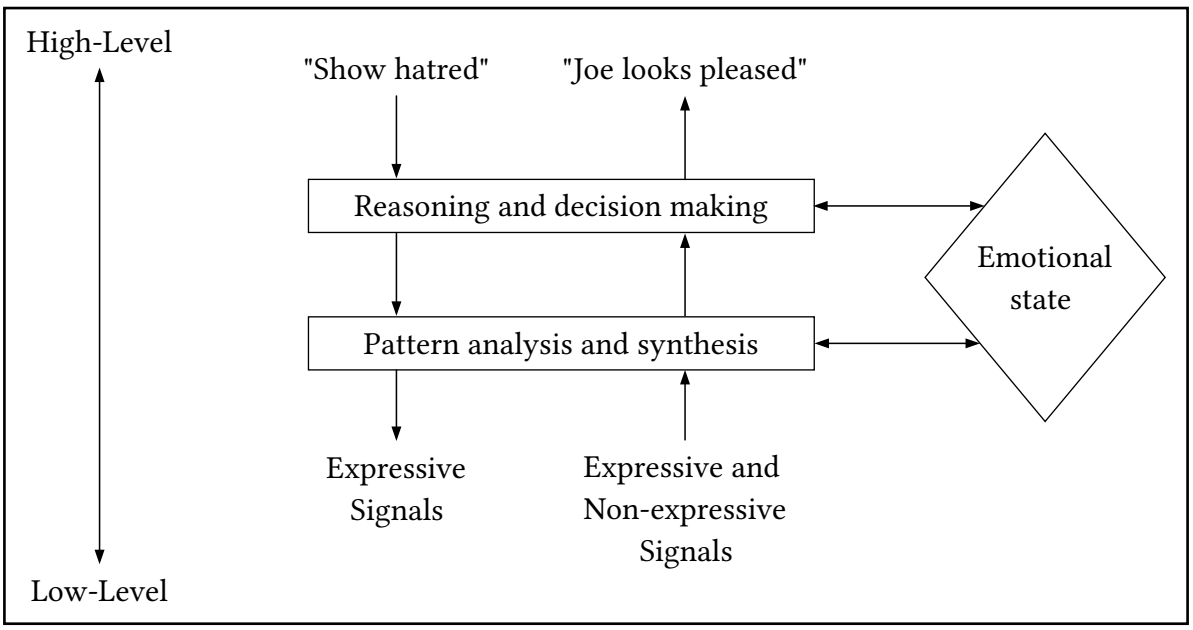


Figure 2.1.: Information flows from both high to low to high in a system that can express, recognize and "have" emotions [2].

High-level information are defined by the semantic of a situation which can be drawn from audiovisual processes for example the command "Show hatred". To show hate a certain emotional state has to be reached and must be reasoned by the situation. As a conclusion low-level information, for instance face expressions, gestures or other syntactic articulations can be observed from the person or system wich executes the given instruction. These patterns are analyized or learned and synthesized by the process before.

On the other direction, these low-level expressive information can be combined to get the semantic of the emotional state. In addition there can be non-expressive data as a valid input. It describes the passive influence on a persons state like the weather or the setting of the environment. Most datasets won't cover this type of information.

The mid-level builds patterns for the given input and combines different low-level signals to synthesize better reasioning for a high-level output. For example a slightly faster speech rate and a tense articulation can be clustered together to get classified as an pleased look.

2.1.1. Emotion Classification

There are different apporaches to classify the semantic high-level features. The most popular classification is the basic-emotion framework by Paul Ekman [3] describing six different emotions: anger, disgust, fear, surprise, sadness and happiness. This arose from different facial expression, which Ekman categorized to the corresponding emotion.

Over the time different frameworks were built on this foundation to introduce for each basic emotion multiple detailled emotions. A common model is the Plutchik wheel of emotion, which is commonly used to classify emotion in text. It offers eight types of basic emotion: joy, trust, fear, surprise and their opposites sadness, disgust, anger and anticipation [4].

Visible in Figure 2.2 each emotion has multiple levels of intensity. For example, the enhanced level for disgust is loathing, a more moderate level is boredom. In total there are 24 different emotions. A variety of sentiments and their intensitys lead to a proper illustration of the real value.

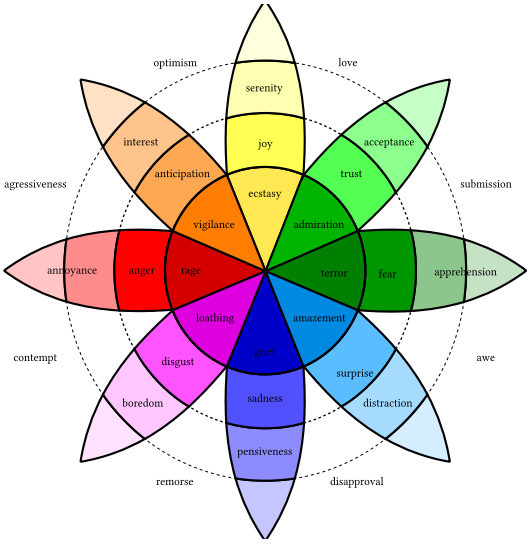


Figure 2.2.: Plutchik Wheel of Emotion [5].

Although there will be a loss of detail, the content of this thesis will focus on the basic emotion from Paul Ekman. The wheel of emotion from Plutchik is suitable for emotion detection in text but most datasets in facial emotion recognition are focused on Ekmans emotion theory. As shown later in this thesis, many multimodal datasets relie on the work from Ekman. It is harder to distinguish all emotion from Plutchik in audio processing and face recognition.

2.1.2. Emotion in Text

The most common approach to classify emotion from natural language text is to classify the semantic content from these sentences. In order to make it possible, two syntactical assumptions are needed.

1. There is a function that is able to map words and sentences to their corresponding emotion.
2. Words and sentences do not change their meaning over time.

The first assumption can be approached by training a deep neural network with a suitable dataset or a clustering algorithm that measures the similarity between words and sentences to match their corresponding emotion label. The function must be surjective, so a token won't be able to be labeled with more than one emotion at the time. The second assumptions is important for a relieable result in training but it is also the flaw from emotion detection in text. For example the sentence "I'm glad to be here". Expressions like irony, emphasis in the voice or gestures can change the proper meaning of the sentence. It may can be interpreted as happiness or as disgust. To take this into consideration further observations of other expressive and non-expressive signals are needed.

2.1.3. Emotion in Speech

Emotion in audio is in contrast to the classification in text mainly focused on the syntax from the recording. The actual signal input is represented as a waveform and sinoid functions. A single wave is defined by

$$g(t) := A \sin(2\pi(\omega t - \varphi)) \tag{1}$$

A complete waveform as presented in Figure 2.3 and Figure 2.4 consists of many different sinoid functions which are added and concatinated together [6].

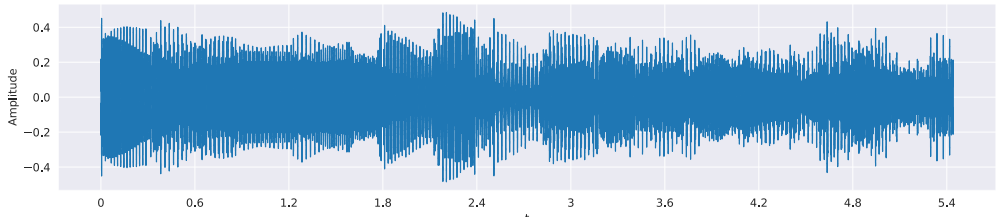


Figure 2.3.: Waveform of the first seconds of a audio recording

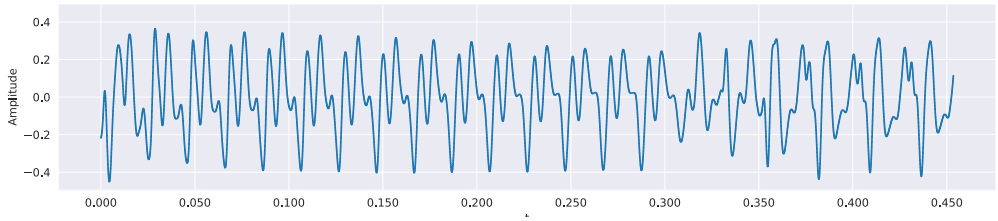


Figure 2.4.: Enlargement of the first half second from Figure 2.3

Frequency (ω) measures the pitch from the audio and especially from the voice. As shown in Table 2.1 multiple features can be extracted from the frequency respectively the pitch. From a measured neutral foundation changes like an average pitch, the range or other changes can be observed. For example a person which speaks in a very much higher pitch, with a wide variablity in the range and changes the frequency aprupt on stressed syllables, then this person feels most likely anger.

Amplitude (A) shows the intensity and the loudness of the given audio. Furthermore it is also an important feature and easy to extract from speech with the help of an existing neutral foundation.

Phase (φ) shifts $g(t)$ along the x-dimension. This parameter is not important in the feature measurement but has a significant role in audio decomposition which is described later.

Extracting features like speech rate, articulation or voice quality more observation is needed. Speech rate can be measured with words per second, sentences per minute or other similar metrics. Normally when looking at waveforms the intermissions betweens sentences or words can get retrieved. A higher rate is equivalent to more observed gaps in the same time of a normal speech speed. Articulation and voice quaility is hard to track without a semantic knowl-

edge. Therefore a combination of the already presented features is needed which can be done by pattern matching and comparison to the neutral foundation.

Emotion / Feature	Fear	Anger	Sadness	Happiness	Disgust
Speech Rate	much faster	slightly faster	slightly slower	faster or slower	very much slower
Pitch Average	very much higher	very much higher	slightly lower	much higher	very much lower
Pitch Range	much wider	much wider	slightly narrower	much wider	slightly wider
Intensity	normal	higher	lower	higher	lower
Voice Quality	irregular voicing	breathy chest tone	resonant	breathy blaring	grubled chest tone
Pitch Changes	normal	abrupt on stressed syllables	downward inflections	smooth upward inflections	wide downward terminal inflections
Articulation	precise	tense	slurring	normal	normal

Table 2.1.: Features of an audio with correlating emotion [2], [7].

2.1.4. Emotion in Facial Expression

Besides his theory about the six basic emotion, Paul Ekman introduced a system to map different face expressions to their relating sentiments by combining changes in different areas of the expression field. This facial action coding system (FACS) is based on these movements which are called action units. There are 28 main actions which can be mapped to the six main emotion. In the following of this thesis, they will be used as the main features.



Neutral Sad Fear Disgust

Figure 2.5.: Examples with their annoations from the CREMA-dataset [8] which show emotions based on facial expression features.

Visible in Figure 2.5 there are visible combinations of face features are able to represent different emotion. For example to observe disgust, commonly the typical wrinkling of the nose, the contraction of the eyebrows and a lifting of the upper lip can be watched. Additionaly to the movement of the eyebrows, sadness is marked through the downturned corners of the mouth. Thus a face and his expressions can be broken down into seperate regions.

Eyes (upper face): The most action units to recognize emotion within facial expressions are defined by movement near the eyes. Eyebrows are seperated into an inner and outer part for a more detailed classifcation. Furthermore they can be raised or pulled together leadning into four different states. Besides from eyebrows, eyelids are another meaningful part in that area. They can be tensed, raised by opening the eyes wide and lowered by narrowing the eyes. In total this region plays an important role for every basic emotion by Ekman and a relevant part for the most sentiments from Plutchiks theory.

Nose: This face region only impacts one basic emotion, disgust, shown in Figure 2.5. This is due to the limited range of movement of this part of the head. Thus this section of the face won't differ from the upper part of the face and the lower part described above due to its low occurrence.

Mouth (lower face): The cheeks and the chin are also correlated to this area for the following work in this thesis due to their tight connection to the lips and mouth by means of muscles. This leads to a total of 16 different action units due to the numerous possibilities, e.g. deforming the lips, opening and closing the mouth and the resulting movements in this area.

In his research Ekman defined each action unit and their corresponding emotion in detail which can be looked up in his book about the basic FACS System [9].

Furthermore in 2002 [10] he updated his idea of the facial expression system and integrated more action units adding up to more than 40 possible features in total. This includes tilting and rotating the head and more details on the upper face like winking or blinking with the eyes.

In addition a closer look at the various intensitys of an action are presented and classified into 5 levels: trace (A), slight (B), marked or pronounced (C), severe or extreme (D) and maximum (E). Most common are slight or marked and pronounced, while the more intens intensities are visible in films or other acted illustrations. Maximum intensness is the most rare but can be find in theater or generally live action plays. The different letters are used as an indicator together with the numerated action unit for a better description and overwiev of all possible combinations.

2.2. Classification Approaches

Over time many approaches to detect and classify emotion have been developed. There are severeral supervised architectures and unsupervised techniques which represent the current state of the art for emotion detection in text, audio and video.

2.2.1. Classification in Text

To classify emotions in text, several preprocessing is necessary due to the lack of numerical values. In the following some ideas for preprocessing are presented. Furhtermore, a look at the various classification methods will be taken. Here a special focus on the supervised methods is applied due to better comparability of already existing and pretrained deep learning models.

Word Embedding and Tokenization

The most important part of all classification approaches is called word embedding. It represents words as real-valued vectors [11]. It shoud use syntactic but also the semantic meaning of the words to get a useful vector. This technique is commonly used in natural language processing like information retrieval in text.

A naive approach to generate such an embedding function would be a one-hot vector encoding [12]. It represents every word in a space as an different vector. For example the sentence:

The tree is green.

will be first tokenized into a set of words V.

$$V = \{\textit{the}, \textit{tree}, \textit{is}, \textit{green}\}$$

The size of the set $|V| = 4$. As a result a mapping $w \in V$ to a $|V|$-dimensional vector is needed.

$$A : V \to \mathbb{R}^{|V|}$$

$w_i \in V$	$A(w_i)$	$w_i \in V$	$A(w_i)$	$w_i \in V$	$A(w_i)$	$w_i \in V$	$A(w_i)$
the	$\begin{pmatrix}1\\0\\0\\0\end{pmatrix}$	tree	$\begin{pmatrix}0\\1\\0\\0\end{pmatrix}$	is	$\begin{pmatrix}0\\0\\1\\0\end{pmatrix}$	green	$\begin{pmatrix}0\\0\\0\\1\end{pmatrix}$

Table 2.2.: Possible one-hot vector mapping for the sentence “The tree is green.”

To measure the similarity the cosine similarity is often used [12], [13]. Similar words or synonyms should have a smaller angle between each other, than unsimilar words.

$$\text{sim}(w_1, w_2) = \cos(\theta) = \frac{w_1 \cdot w_2}{\|w_1\| \, \|w_2\|}$$

While the one-hot vector embeds it word by word only without any context, there won't be any similarity or dissimilarity between the words. This is also the reason why simple one-hot enocding is less used in practise. A semantic consideration will be more useful.

The Continous Bag-of-Words Model, shortly CBOW is capable to take the context of a word as an input to calculate an embedded vector with emphasis on syntax and semantic. It consists of a neural network which mainly consists of three layers: the input layer, a hidden layer and the output layer [14].

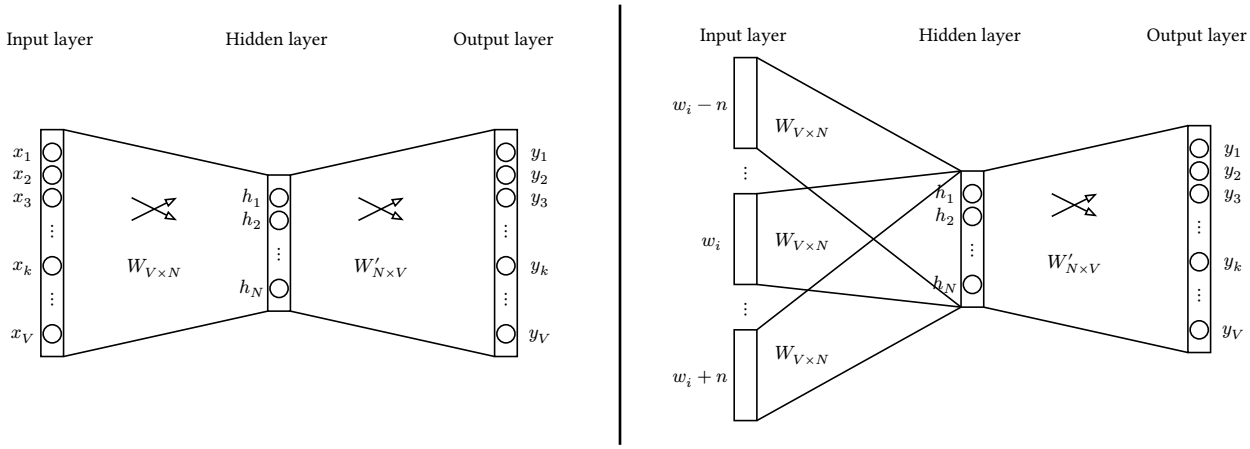


Figure 2.6.: Left: Simple CBOW-architecture for a single word; Right: CBOW-architecture for multiple words. Words have length k and are padded to size V to match the dimension [12].

The architecture shown on the right side of Figure 2.6 takes one one-hot vector $A(w_i)$ as an input expression. To get the context of the inputted term, phrases before and after are needed. Therefore an addition to the architecture is necessary.

With unspuervised learning the model on the right of Figure 6 is able to generate a new vector which represents the context of a word. Now, under consideration of the cosine similarity, words with a similar context have a quite smaller angle compared to without any connection in semantics. In both architectures, the hidden layer acts like an arithmetic operator which combines the context to the outputted one-hot vector.

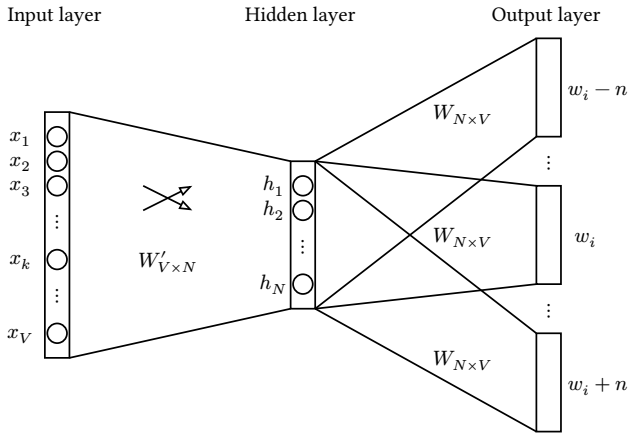


Figure 2.7.: Skip-Gram Model. All words have size V (after padding) [15].

The Skip-Gram Model calculates an embedding in reverse from the idea of the CBOW-Model. As the architecture shows in Figure 2.7 it tries to calculate the surrounding words with a given target term [16]. Therefore a training like the CBOW with contextual phrases is needed before it can work like a semantical decoder. Both architectures are commonly used in different pretrained word embedding networks, like Word2Vec from Google [15], [17].

Fast WordPiece Tokenization is an approach to tokenize sentences and words in an efficient way by splitting words into smaller subwords. It is used in a language understanding transformer model from Google which is called Bert [18].

The basic problem from other tokenization methods like the normal WordPiece Tokenization starts in the vocabulary tokenization in tries. Trie memory is a paradigma which is able to store strings efficiently where each edge of the data structure stores phonemes. Traversing a path consisting of multiple edges results in a word. As a result the structure does not have to store reduntant parts of words [19].

The tokenization has to search for the longest token at a position. Therefore it starts from the shortest substring and iterates over the input text, while trie matching to match the longest prefix which matches a vocabulary token [20]. If the trie matching fails at the next character, the algorithm has to involve backtracking because the longest vocabulary token may be several characters back. This results into a time complexity of $O(nm)$ where m is the maximum vocabulary token length and n the input length.

Fast WordPiece Tokenization introduces so called failure links and pops. Failure pops are a list of longest-matching prefix tokens of a node in the trie, which can be removed in order until a failure link node is reached. The failure node is able to represent the missing suffix. This whole algorithm called LinMaxMatch has a time complexity of only $O(n)$.

Supervised Classification Approaches

With a given word embedding and tokenization supervised learning approaches are able to recognize emotion in text. Therefore an annotated dataset is needed, which has a relieable mapping from sentence or word to the corresponding emotion.

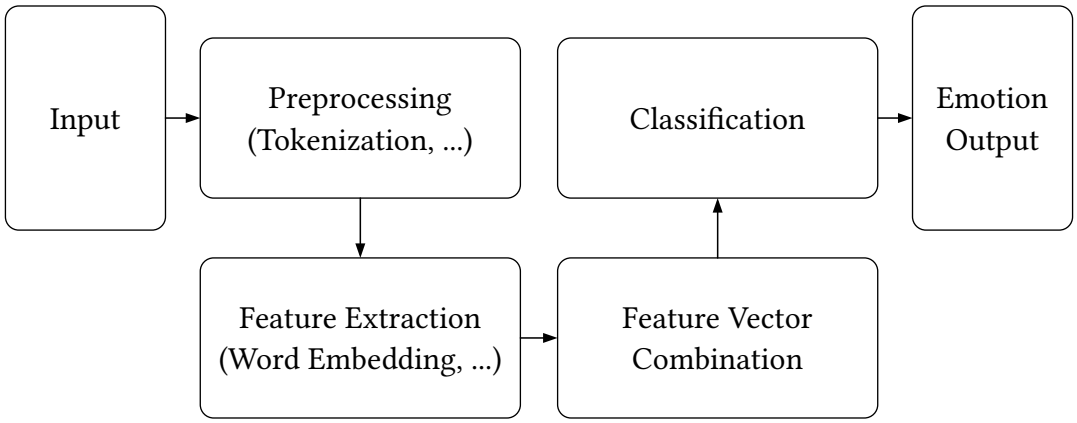


Figure 2.8.: Deep Learning Assisted Semantic Text Analysis

Most models are using the architecture paradigma shown in Figure 2.8 . They mostly differ in the step of combining the feature vectors ant the type of classification because it offers the biggest room to maneuver. A preprocessing and feature extraction is always necessary. Sometimes a WordNet [21] is integrated to get a better overview about synonyms or to paraphrase words.

Convolutional and Recurrent Neural Networks (CNN/RNN) are an effective approach on sentiment analysis of text [22]. Convolution consists of a kernel-funciton which downscales the input but keeps the important features. This technique is called Pooling and brings several benefits like less memory usage, less overfitting in training and a faster computation time. Herefore multiple words or sentences of the space V are combined to region matrices. On each region L convolutional filters are applied to lern local n-gram features.

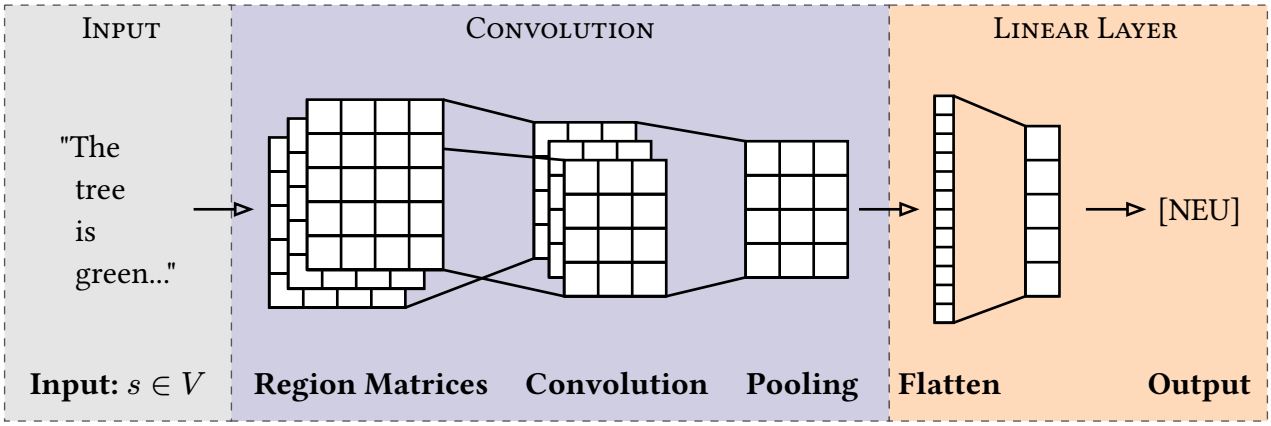


Figure 2.9.: Simple Convolution based Neural Network to predict an emotion from a given subset of sentences

Figure 2.9 shows such network with the convolutional part. In common architectures multiple convolution and pooling layers are stacked together with an activation function like ReLu in between of each block which improves the output by setting constraints to the convolution outputs. The constraint which is effected by ReLu is the prohibition of negative values by the function $f(x) = \max(0, x)$. This subsequency of the convolution blocks leads to a hierarchical decomposition of the given embedded region matrix. As a result multiple features can be found from each block and each convolution kernel which might result in a higher accuracy. At the end an ordinary decoder in form of a simple linear layer with an activation function like SoftMax is utilized to calculate the resulting emotion from the pooled and flattened vector. Normalization like layer norm or batch norm between each layer can help to ensure better propagation and reduce numerical instability [23].

Another optimizations to fundamental neural network architectures are gating mechanisms from reccurent networks and the bi-directional flow in the model forwarding architectures. They introduce new types of storage cells which allows to store long term features [24]. In the context of emotion detection in text, it might store important context from sentences and words before to differ between fill-words like “this”, “with” and emotionally loaded words. Commonly used are long short term memory cells (LSTM) [25] and gated recurrent units (GRU) [26] which both try to prevent gradient vanishing which is a big problem in recurrent networks. It occurs while propagating backwards through time. Due to many multiplication steps, the gradient converges to 0. These two architectures differ mostly in their structure than in their functionality. Additionaly to the GRU cell, an LSTM has two more control gates which gives more possibilieys to constrain and rule the memory inside the structure. In both cases they can improve learning features from context and time series.

Transormers are an deep-learning architecture which is based on multi-head attention.

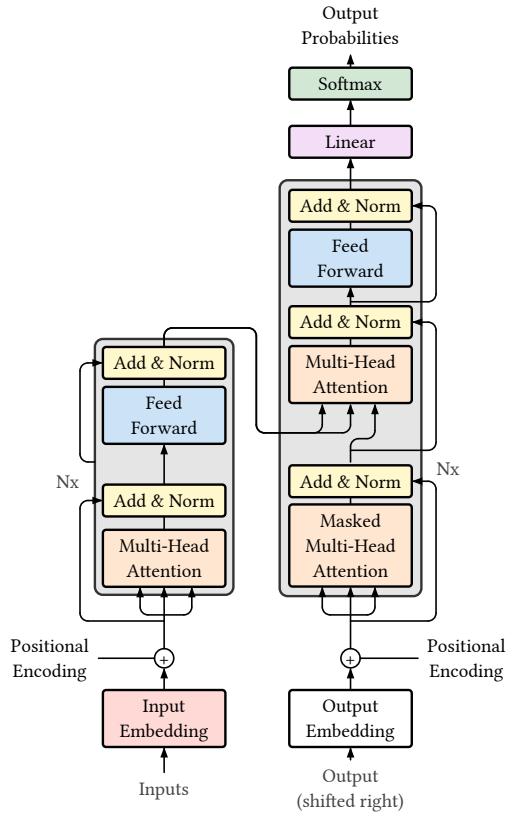


Figure 2.10.: Common transformer model architecture with multi-head attention [27].

The basic idea of attention solves a huge bottleneck in common neural network architectures because fixed input vector sizes may result in a large overhead. In the example of natural language processing this could be a result from very large sentences. Attention mechanisms try to solve this issue by focusing the input on to specific parts e.g. training the input weights on important words or subsets from the given sentence.

The resulting transformers are currently setting a milestone in natural language processing and providing the fundaments for ChatGPT from OpenAI [28] or other text understanding models, because of their scalability due to the attention mechanisms and their relieable results. Figure 2.10 shows such a transformer architecture. An input sentence will be fed into the model where it first gets positional encoded thus the position and the frequency for every word in the sentence is stored.

Now the input vectors are passing through the first multi-head attention layer to lay the focus on important parts of the input. Neverless the actual input and the weighted part are added and normalized together which results in the original sentence but with tagged highlights due to the attention. To learn the features for the output token, the input traverses through a feed forward network which consists of basic uni-directional layers. Likewise the input and the output are matched together with a residual connection. The same proceding is repeated but with an autoregressive addition of already calculated output. But in contrast to the regular multi-head attention a masking in this process is introduced which sets the weight for unknown or respectively future tokens to negative values thus the model can learn realistic predictions only with already known tokens. The output of the transformer architecture are probabilities for the next state. The already mentioned autoregressive implementation gives the possibilities to predict a long sequence of tokens.

In addition to the approaches with neural network architectures like transformers, other supervised methods like support vector machines, decision trees or naive bayes are used. These aren't covered in this thesis as it exceeds the scope of this work and are a possible part for future work.

In summary supervised methods follow the basic principle to learn features along a given target. Therefore enough training data is needed with the additional annotated labels.

At the end even the differentiation between convolution, reccurent or transformer networks won't make any noticible changes in measurements if the architectures and models are large enough [29]. In literature this "bitter lesson" [30] is researched step by step and also shown for language models, convolution nets and transformer.

"Scaling Laws for Neural Network" [31] shows the improvement of the loss, when a model has many parameters, much computation time in training and a large dataset. Therefore this thesis will consider the transformer models as a focal point for later evaluation, because there are many large pretrained language models available which are able to recognize emotion and are trained and developed to satisfy almost all three factors. Language modeling performance improves smoothly as the model size, datasetset size, and amount of compute used for training is increased. For optimal performance all three factors must be scaled up in tandem.

Unsupervised Classification Approaches

Many datasets do not include annotated targets or subsets of the data can't be used as a prediction set. In unsupervised learning, the algorithms are trying to extract patterns from this data which are mostly similarities or differences in the training set.

A common unsupervised approach in natural language processing for extracting keywords or similarieties from text is called Latent Semantic Analysis.

Latent Semantic Analysis (LSA) [32] is an information retrieval method in text based on singular value decomposition. The algorithm first transforms every word from a sentence into a matrix M. This matrix can be dismantled into a diagonal matrix S and two orthogonal matrices U, V through

$$M = U \cdot S \cdot V^T \tag{2}$$

S contains singular values which can be easily converted to eigenvalues. These are used to process a query on the vector space to find subsets of M which fullfills the query by applying the cosine similarity. This whole method is quite expensive in computation time because of the necessary decomposition and dimension reductions [33].

There are many more unsupervised approaches for emotion detection and information retrieval in text [34], [35] but they won't be part of the following comparisons and benchmarks s as they also exceed the scope of this thesis. Furthermore, no significant improves and diffferences are expectet in comparison to the presented unsupervised approaches and also the shown supervised classification methods due to the scaling laws for neural networks.

2.2.2. Classification in Audio

Similar to classification in text as shown in Chapter 2.2.1 there are some approaches to suppervised and unsupervised methods out there. First of all an encoding for the audio is necessary to ensure the extraction of the necessary features.

Fourier Transformation

To break up the audio signal into multiple atomic sinoid functions, a fourier transformation and analysis can be applied. For each frequency $\omega \in \mathbb{R}$ from the given signal, the transformation yields a coefficient d_ω and a phase φ_ω from which a sinusodial oscillation can be retrieved. Looking back at the equation from Chapter 2.1.3 a normalization with regard to their amplitude is collected by assuming $A = \sqrt{2}$.

$$g(t) := A \sin(2\pi(\omega t - \varphi)) \tag{3}$$

For each frequency and phase the normalized function $\cos_{\omega,\varphi}(t) : \mathbb{R} \to \mathbb{R}$ is given by:

$$\cos_{\omega,\varphi}(t) = \sqrt{2} \cos(2\pi(\omega t - \varphi)) \tag{4}$$

Due to the idea of approximating the original audio function with sinoid waves, a first approach to the fourier transformation can be obtained by maximizing the similarity between both signals.

$$\begin{aligned} d_\omega &:= \max_{\varphi \in [0,1)} \left(\int_{t \in \mathbb{R}} f(t) \cos_{\omega,\varphi}(t) \, \mathrm{d}t \right) \\ \varphi_\omega &:= \operatorname{argmax}_{\varphi \in [0,1)} \left(\int_{t \in \mathbb{R}} f(t) \cos_{\omega,\varphi}(t) \, \mathrm{d}t \right) \end{aligned} \tag{5}$$

Bringing this into the context with polar coordinates and the additional idea to store both values d_w and φ_ω into a single complex value. This results in the common fourier transformation form given by:

$$\begin{aligned} \hat{f}(\omega) &= \int_{t \in \mathbb{R}} f(t) \exp(-2\pi i \omega t) \, \mathrm{d}t \\ &= \int_{t \in \mathbb{R}} f(t) \cos(-2\pi\omega t) \, \mathrm{d}t + i \int_{t \in \mathbb{R}} f(t) \sin(-2\pi\omega t) \, \mathrm{d}t \end{aligned} \tag{6}$$

These equations are however only suitable for non-discrete signal-processing and must be adaptet to adaptet to a discrete fourier transformation. A sampling period T is introduced to sample the signal in mutiple time steps. An approximation for $\hat{f}(\omega)$ is obtained

$$\hat{f}(\omega) \approx \sum_{n \in \mathbb{Z}} T(f(nT)) \exp(-2\pi i \omega n) \tag{7}$$

with a time complexity of $O(N^2)$ where N notates the length from the given signal [6]. Thus N can get quite large some approvements are implemented. For example the fast fourier transformation with a complexity of $O(N \log N)$, uses a divide and conquer algorithm to split the signal, calculates the fourier transformation and reassembles it [36]. Also common is the short time fourier transformation which divides the audio into windows with a given size n and a hop size to determine the step size of the sliding window [6], [37].

Supervised Classification Approaches

Analog to Chapter 2.2.1 the supervised classification approaches follow the same structure. Recurrent- and convolution-based networks are the most common approach [38], [39] to classify features from audio, after retrieving the fourier representation for every signal and can also be applied to emotion. Continuing from the already presented fourier transformation and analysis, these supervised classification methods refer to the feature extraction from spectograms [40], [41] which is the visual representation of the given signal and a lightweight implementation for these networks.

Meanwhile the first transformer based audio classification are presented [42]. The idea is an amalgamation of the idea from convolution architectures and the common transformer model.

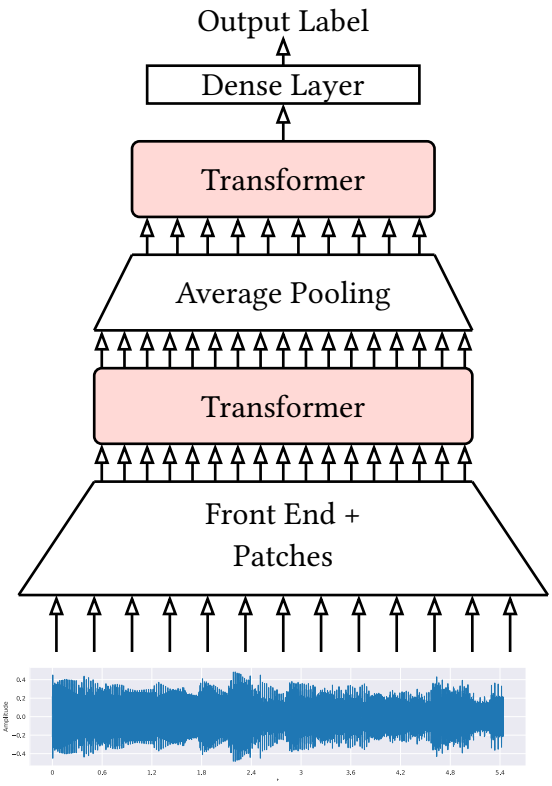


Figure 2.11.: Audio transformer model

Therefore it inputs the raw audio data without any preprocessing besides from padding to the right length. As shown in Figure 2.11 the model first divides the signal into multiple fix-length patches to feed them into a front-end layer. The patches describe very short sequences of the audio and show the resolution in which the work will subsequently be carried out. It consists of some dense layers which processes the raw audio to multiple feature vectors. These are suitable for learning in the common transformer architecture. These are chained together with average pooling layer, giving the convolution like downscaling and compressing effect. This step can be repeated multiple times until the required output shape is nearly matched. At the end a decoder layer in form of some dense layers are applied to predict the resulting label. Equivalent to the already introduced convolution networks, activation functions like ReLu or softmax can be interposed between all existing layers for more stability.

Nonetheless the bitter lesson from the previous chapter also applies here. Therefore and because of the already large scope of this thesis, further considerations about other supervised approaches like SVM or decision trees, are not made.

Unsupervised Classification Approaches

Unfortunatly unsupervised audio information retrieval approaches are quite rare. There are a few methods which uses k-means to learn from local audio features [43]. Thus they are not really suitable for emotion detection due to their low capability of detecting global features. But in spite of that there are some methods to collect local features which can assist the supervised deep learning method [44].

2.2.3. Classification in Images and Videos

Relating to the previous classification methods the foundation for emotion detection in images and videos is quite similar. However there is many research to optimize the supervised and usupervised strategies regarding the input size and computation time. Furthermore video as interesting advantages and disadvantages towards images which is necessary for the following methods and optimizations. The disadvantages comes with the additional dimension, time, which is introduced through videos as a time series while they can be dissected into single images.

Supervised Classification Approaches

As always convolution models are the most common architectures for deep learning on images and videos [45]. While images can be flattened to an one dimensional vector to process it with simple convolution filters, layers or even with recurrent and simple linear models, video processing and learning arrives with a bigger challenge. Fortanatly kernels for convolution are able to have n-dimensional shapes though getting expensive in computation and memory consumption. The same issue occurs naturally with video transformers [46]. These follow anaolg to the already presented architectures in text and audio classification as they are transfering the researched benefits from transformer models. Nevertheless they scale quadratically in respect to the input length. Thus further optimizations on the input are helpful and even more necessary.

Slicing offers different options to reduce the input size from the video with a variety of selection methods [47].

- **Time Exclusive Slicing:** As the most intuitiv approach, time exclusive slicing picks one or more single frames out of the given video, resulting into a more compact version like a flip book. This selection can be done randomly or with a fixed selection of the frame indices [48].
- **Spatial Exclusive Slicing:** This method divides each frame into a grid of spatial patches . Then it chooses one of the grids (mostly the center) and picks the same patch over the full time dimension.
- **Cubicles:** Cubilcles or tubelets [49] are extracted by choosing spatial patches in combination with its temporal neighbours. This is similar to spatial exclusive slicing but with more variety in the choosen grid positions.
- **Shards:** This method [50] is a combination of spatial exclusive slicing, cubicles and time exlusive slicing, where from random or fixed indexed frames the spatial patches are choosen.

Overall they reduce the input size of the input video to a fraction of the original size on the basis of the assumption that there are unnecessary parts in each frame like the corners and that videos with more frame per seconds includes more redundancy respectively too small changes between each time step. Furthermore the basic spatial exclusive slicing is not well suited due to the lack of removing the de-

scribed redundancy. In general spatial patching can be challenging, if the person in the video is moving or the main actors are on different position in each data sample.

Further tries for input optimizations regarding to the idea of tracking movements are researched like utilizing temporal difference images [47]. It follows the assumption that deep learning architectures are learning the features only from the moving part of the face and that non moving parts are unnecessary. Intuitive according to Ekman and the presented action units from Chapter 2.1.4 this hypothesis has a good foundation. With time exclusive slicing they picked frames both randomly and with a fixed distance between the time steps and subtraced the frames from each other to create a temporal difference image for each video. Each of them shows the moving action units which were present in the data sample before. Thus it shows the action units which are normally learned by the first layers from an architecture.

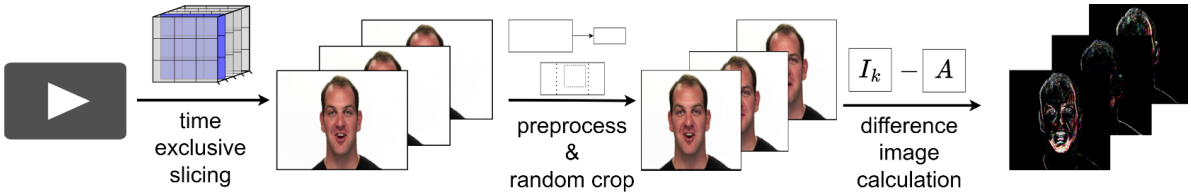


Figure 2.12.: An overwiev of the preprocessing pipeline [47].

This is shown by Figure 2.12 where the resulting temporal image shows only the moving parts from the video. Thus it may save some computation time but mostly it is able to save memory cost due to the reduction from 30-120 frames per input to only one temporal frame.

A similar but yet different approach is called optical flow [51], where for each video the moving parts and how they changes are calculated and represented as multiple feature vectors. Here a video is represented mathematically by $I(x, y, t)$ where t denotes the time dimension as the single frames. The coordinates x and y belong to points defined in advance at which the movement is then tracked. These can be for instance the corners of the mouth, parts of the eyebrows or the eyelids. More points can be used to get a more precise resolution but also a larger input sequence. An appropiate amount would be between 20 and 30 to capture all 26 basic action units and some more details. The resulting optical flow while tracking is denoted by

$$\frac{\mathrm{d}t}{\mathrm{d}x} = I_x u + I_y v + I_t = 0 \tag{8}$$

where (u, v) representing the direction where the pixel at (x, y) is moving to. The resulting flow matrices can be used for a convolution deep learning architecture to determine the corresponding emotion. All in all there are these approaches to reduce the memory and computation time problem. If these aspects can be neglected a simple 3D convolution network will cover the time dimension and can be adapted to the structure of a video [52].

Unsupervised Classification Approaches

The challenge for unsupervised classification is the feature extraction from the video which needs some kind of tracking like optical flow. Thus it has to determine which action unit is activated leading to a combination as presented in Chapter 2.1.4. The already shown preprocessing possibilities from supervised approaches are visualizing a first look about possible feature extraction, which can also be helpful for unlabeled datasets.

Appearance-based Approaches are following the idea that different facial regions are defined by unique colours and textures. In general it is approached by mapping a coordinate grid absolute to a fronatal picture of a head [53] or relative to fixed marks [54]. Each face region is divided into non-overlapping patches and blocks. Each block is represented as a single feature vector and contains one facial landmark which can be afterwards clustered to an action unit. Thus it observers the moving parts by colour, texture and the position in the frontal face it is sensitive to changes of light and even smaller changes of the head position. Therefore it is a suitable method for testing and methodological improvements but is not quite suitable for real life application due to these sensitivities.

Geometry-based Approaches are not well researched due to extracting and obtaining facial point locations [55]. However somewhat similar to the appearance-based approaches the idea for geometric feature extraction also looks for changes in the static face but more for deplacement of the same point location due to muscular changes. In combination with the approach from before it can result in improved performance independent from lighting and outer conditions.

Motion-based Approaches are based on dense motion rather than the discrete motion from facial landmark points as presented in geometry-based approaches. Image substraction, as one subset, which was already shown during input minimation for supervised approaches can also be used in combination with a combination of basic clustering algorithms like k-means [56]. Also the presented optical flow method is capeable to be used for unsupervised learning adapting it to quadtree decomposition for matching the important moving parts.

In general these feature extraction approaches uses k-means with an alignment to time series. Oftenly used is dynamic time warping, which a distancy metric and useful for time series with different lengths. Thus it has a drawback due to not satisfy the triangle inequality. Further optimizations lead to a method called dynamic time alignnment kernels [57] defined by

$$\tau = \max_{Q} \sum_{c=1}^{l} \frac{1}{n_x + n_y} (q_{1c} - q_{1c-1} + q_{2c} - q_{2c-1}) \kappa_{q_{1c} q_{2c}} \tag{9}$$

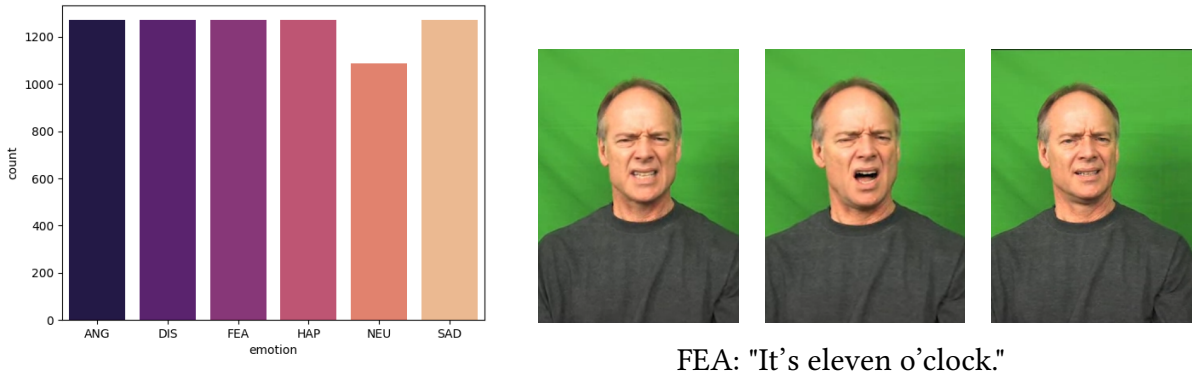
where $\kappa_{q_{1c} q_{2c}}$ represents the kernel similarity between single frames from two video inputs. A possible kernel could be $\kappa_{q_{1c} q_{2c}}(x_i, y_i) = \exp\left(\frac{1}{2\sigma^2} \parallel x_i - y_i \parallel_2^2\right)$ with σ^2 as the average distance to the 10% nearest neighbours [58]. Clustering the features is not limited to k-means but is the most common approach. Thus the research in time series clustering has developed over the last years bringing different metrics and algorithms to the front and can be adapted with little to no effort [59].

2.3. Datasets

For the multimodal analysis and the supervised approaches in learning a dataset consisting of at minimum video with audio is needed. From this state the text transcription can be retrieved during preprocessing. During this thesis, two different datasets are used which fulfill the requierments. The following datasets are based on the Ekman emotion model with some changes.

CREMA-D (Crowd-sourced Emotional Multimodal Actors Dataset) [8] consists of 7.442 clips from 91 different actors with their facial and vocal expression. It covers every of Ekmans emotions: *Anger (ANG), Disgust (DIS), Fear (FEA), Happy (HAP), Neutral (NEU), Sad (SAD)* with the exception of *surprise (SUP)* , because it may can relate to any other emotion with a rapid onset. Additionaly there are a few annotation to the intensity of the sentiment, but it is only a small percentage regarding the amount of data. Therefore it wont't be used for futher evaluation in this thesis.

They recorded twelve different sentence for each type of emotion like "Don't forget a jacket." or "I wonder what this is about." which are semantically neutral connotated. These snippets are between 50 to 80 frames long while it is recorded in 30 frames per second.

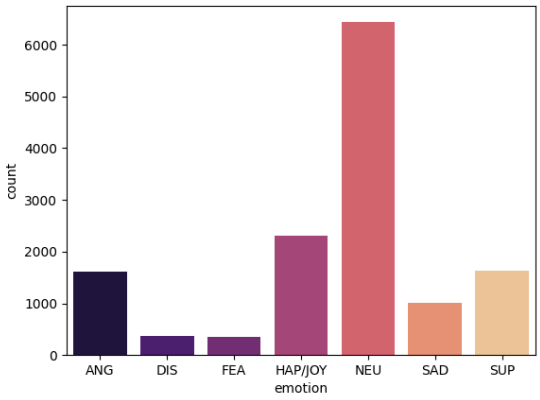


FEA: "It's eleven o'clock."

Figure 2.13.: Distribution and example of the emotion presented by the CREMA dataset

As shown in Figure 2.13 the distribution of labels is uniformly distributed with the exception of neutral. This won't leed to any problem, because the actual sentiments shown in the visuell data are very strongly acted so there are clear features available.

MELD (Multimodal EmotionLines Dataset) [60] is an annotated collection of short sequences from the series "Friends". In difference to the CREMA dataset or most other multimodal datasets for sentiment analysis like IEMOCAP [61] the given data has many multi-party conversation where there are more then one participant active and present in the most parts. This leads to a more complex and challenging feature extraction but approximates more closely the reality.





JOY: "You are a genius." (Chandler)

Figure 2.14.: Distribution and example of the emotion presented by the MELD dataset

Figure 2.14 shows the uneven distribution of the different sentiments, which can lead to an imbalanced training. Nevertheless it is a quite large dataset with 13.708 annotations which includes in addition to the emotion, the character as the speaking person which can be used to analyze the different features depending on gender or age.

Chapter 3.

Related Work

Emotion detection is a research field which has developed much over the time. Also the comparison of multimodal emotion detection approaches and their results crystallized in the last few years. In the following chapter multiple exploration are presented with their idee and the result. Also a comparison of the different approaches will take place mostly while presenting the ideas shown in the works.

3.1. Audio and Video Based

Temporal Multimodal Fusion for Video Emotion Classification in the Wild by Vielzeuf et al. [62] builds a framework on audio and visual features where it tries different combination and different scoring of the features. They use convolution models as a deep learning architecture and the OpenSmile toolkit as an audio feature extractor to retrieve supervised the six emotion from Ekman and in addition the neutral state. The visual information are extracted frame by frame and as a conecutive video whithin two different models. For training they used the Acted Facial Emotion in the Wild dataset [63] which consists of different movie scene and is comparable with the presented MELD dataset. In total they have an accuracy of 58.8% with the combination by a weighted mean but with no information about how they set the weights.

Multimodal mixed Emotion Detection [64] presents a similar method and implementation to the work presented before. Thus they create features from hand gestures and body movement aswell which are then forwarded into a support vector machine. To extract information about the visual and audio data they used premade frameworks like OpenEar similar to OpenSmile and FaceAPI which allows better tracking for faces and head rotations. Additionaly they used an infrared sensor from Microsoft for the gestures and moving parts of the body. Furthermore they classified the six Ekman emotion like the paper before but without any neutral emotion. As results they presented an accuracy of 77.5% while using data from the hand gestures, 65.2% for the body modality and 94.3% for using information from the head region. In total by combining all features they got an accuracy of 96.6% which is quite promising. Thus they are using the IEMOCAP database which is more comparable with the CREMAD database and brings more acted emotion into training than a more natural dataset would do which was used in the work before. Also they did not include any feature combination with weights or other methods. The high difference in the final accuracy shows the difficulty with natural moving parts or moving cammera perspectives like shown in movie scenes instead of static positioning and overacted emotion.

3.2. Audio, Video and Text Based

Multi-level Feature Optimization and Multimodal Contextual Fusion for Sentiment Analysis and Emotion Classification [65] is using text, audio and video as input, extracting the features with evolutionary computing based models like GA or GGA [66]. They propose a multi layer feature extraction which propagates the raw input through different layers. The first subsets of information and patterns is extracted by unimodel GAs and GGAs, and then forwared into an recurrent LSTM network to retrieve more contextual input. This procedure will be repeated multiple times and between each step a fully concatenation of all features is applied and a split from unimodal to a trimodal architecture. Fot training and testing they used the IEMOCAP database as a static dataset and the CMU-MOSI data. This set consists of opintion videos from YouTube splitted up into multiple emotionally affected segments making it compareable with the MELD or the WILD database. As shown in Table 3.1 they got quit relieable results showing that a combination of the multimodal features can improve the accuracy. In single feature usage they got an score of 69.49% for video, 77.28% in text and 68.28% while using the IEMOCAP dataset and similar results for the CMU-MOSI model.

Modality	Accuracy CMU-MOSI	Accuracy IEMOCAP
T+A	77.06%	78.85%
T+V	76.65%	79.12%
A+V	70.00%	65.44
T+A+V	80.38%	79.52%

Table 3.1.: Comparison of results. A = Audio, V = Video, T = Text.

To combine the features they used the contextual fusion where the model trains the optimal weights for the composition. Furthermore it shows that there won't be much improvement between text and audio in turn of the same features combined with video features. Thus it presents a relieable architecture for both different types of datasets.

Multimodal Sentiment Analysis using Hierarchical Fusion with Context Modeling [67] follows a slightly similar approach like the work before. Instead of GGAs for feature extraction they used basic convolution networks trained with the IEMOCAP and CMU-MOSI databases. Thus for audio OpenSmile was used as a pretrained processor. They present an mostly identical feature combination strategie where a fully connected layer and gated recurrent units for contextual learning to combine the features and learn the optimal weights. Overall they achieved a little less accuracy than the feature extraction with generative models but for text, video and audio in combination they got a score of 80.0%. Audio and video achieved 58.8%. Bimodal combinations with text got an accuracy of round 79% leading aswell to the insight that there is not much room for improvement in comparison to the trimodal result. For the single model evaluation, text performed the best with an accuracy of 76.5%. The other two models bring about 50% to the evaluation which is quite low in comparsion to the already presented methods.

Seq2Seq2Sentiment: Multimodal Sequence to Sequence Models for Sentiment Analysis researched by Pham et al. [68] presents architecture based on a special class from recurrent networks. The basic idea is to first train a sequence-to-sequence model for two modalities and afterwards a combinition of the already trained model and the third modal to the final result. Therefore they use the CMU-MOSI dataset and for feature extraction they build on different pre-trained encoder. For text input a GloVE [69] which represents his own idea of encoding besides skip-gram or bag-of-word models thus they used an already pre-trained one. COVAREP [70] was used to extract multiple important features and feature represenation from the audio signals and for videos an motion-based representation extraction method is proposed. Altogether they didn't measured the accuracy for single modal training due to the structure of architecture. Thus they got an accuracy of 75% for all three modalities in combination. Interesting to note are the accuracys for the bimodal detections. Text in combination with video got 78%. The other two modal permutations lay at 44% for text with audio and 55% for audio and video. They discussed these results with the reasioning that the model architecture did poor training on video and audio features and recommend other encoding mechanisms and a bit different model architecture basing on sequence-to-sequence for future work.

Tensor Fusion Network for Multimodal Sentiment Analysis [71] presents the method of tensor fusion for the necessary feature combination in context of the multimodility. It describes an untrainable algorithm which attemps to combine multiple tensors by a reduction to a single tensor. This results in an interplay between a recurrent and LSTM based model for each modality and the proposed tensor fusion layer. Only for text input they showed an own method to extract the necessary features also by combining LSTM cells and linear layers. Like the other works before they set their foundation for audio extraction on pretrained models like COVAREP but they combined multiple processors for different features as peak slope parameters or other high level audio information. For video they also refer to FACET as the facial expression analysis framework able to extract up to 20 action units. On to the tensor fusion layer some fully connected deep layers will learn from the combination tensor to predict the emotion. Overall they reached with their unimodal netowrk an accuracy of 77.1%. Also here the baselines representing every single modal are bringing a high accuracy for text processing by 74.8% and lower precision for visual and acustic features with round about 66%. It outperforms the basic convolution network proposed by the first work and reaches good performance in relation to all other presented works thus being a more rare approach while using an untrainable feature combination.

2 more paper?

All in all the current works in the field of multimodal emotion detection are not showing much difference to each other. As shown in the following Table 2.1 the main difference are laying in the basic training architecture to train the feature combination. Thus there are mostly some kind of reccurent architectures deposited to keep track and of context bringing an important focus. Furthermore most of the current works are bringing some indistinguishable accuracys but showing that a bimodal combination of text and audio brings a nearly good result in comparison to the unimodal architectures. Also the usage of pretrained processing layers and feature extraction like FACET, GloVE and OpenSmile is highlighting which servers a huge reduction in computation and necessary preprocessing time. It should be noted that there is no research about the multimodal emotion detection featuring the leading transformer architectures which could be found.

Author	Year	Dataset	Architecture	Fusion Method	Accuracy Bimodal	Accuracy Trimodal
Vielzeuf et al.	2017	WILD	Convolution	Weighted Mean	58.8% (A + V)	-
Majumder et al.	2018	IEMOCAP	Support Vector Machine	Linear Layer	96.6% (A + V)	-
Huddar et al.	2020	CMU-MOSI	GGA, GA, Recurrent LSTM	Contextual Fusion	77.06% (T + A) 76.65% (T + V) 70.00% (A + V)	80.83%
Huddar et al.	2020	IEMOCAP	GGA, GA, Recurrent LSTM	Contextual Fusion	78.85% (T + A) 79.12% (T + V) 65.44% (A + V)	79.52%
Majumder et al.	2018	IEMOCAP, CMU-MOSI	Convolution	Recurrent Networks	77.30% (T + A) 77.80% (T + V) 58.80% (A + V)	80.00%
Pham et al.	2018	CMU-MOSI	Seq2Seq	Hierarchical Fusion	44.00% (T + A) 78.00% (T + V) 55.00% (A + V)	75.00%
Zadeh et al.	2017	CMU-MOSI	LSTM	Tensor Fusion, Linear Layer	-	77.10%

Table 3.2.: Overwiev of the compared works showing the bimodal and trimodal results. A = Audio, V = Video, T = Text

Chapter 4.

Implementation

- Einteilung in drei Multimodal
- alles andere: Unterkapitel
- konzeptuelles Vorgehen beschreiben (Flow Chart)

4.1.

4.2. Prompt Engineering for LLM

4.3. Model Architectures

4.4. Training

4.5. Scoring and Weighting

Bibliography

[1] W. James, "What is an Emotion?," *Mind*, vol. 9, no. 34, pp. 188–205, 1884, Accessed: Mar. 11, 2024. [Online]. Available: http://www.jstor.org/stable/2246769

[2] R. W. Picard, "Affective computing." MIT Press, Cambridge, Mass. [u.a.], 1997.

[3] K. R. Scherer and P. Ekman, Eds., *Approaches To Emotion*, 1st ed. Psychology Press, 1984.

[4] P. Kumar and M. Vardhan, "PWEBSA: Twitter sentiment analysis by combining Plutchik wheel of emotion and word embedding," *International Journal of Information Technology*, vol. 14, no. 1, pp. 69–77, Feb. 2022.

[5] D. Chafale and A. Pimpalkar, "Review on Developing Corpora for Sentiment Analysis Using Plutchik's Wheel of Emotions with Fuzzy Logic," *International Journal of Computer Sciences and Engineering (IJCSE)*, vol. 2, pp. 14–18, 2014.

[6] M. Müller, *Fundamentals of music processing: Audio, analysis, algorithms, applications*, vol. 5. Springer, 2015.

[7] I. R. Murray and J. L. Arnott, "Applying an analysis of acted vocal emotions to improve the simulation of synthetic speech," *Computer Speech & Language*, vol. 22, no. 2, pp. 107–129, 2008, [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0885230807000393

[8] H. Cao, D. G. Cooper, M. K. Keutmann, R. C. Gur, A. Nenkova, and R. Verma, "Crema-d: Crowd-sourced emotional multimodal actors dataset," *IEEE transactions on affective computing*, vol. 5, no. 4, pp. 377–390, 2014.

[9] E. Friesen and P. Ekman, "Facial action coding system: a technique for the measurement of facial movement," *Palo Alto*, vol. 3, no. 2, p. 5–6, 1978.

[10] J. C. Hager, P. Ekman, and W. V. Friesen, "Facial action coding system," *Salt Lake City, UT: A Human Face*, p. 8–9, 2002.

[11] B. Wang, A. Wang, F. Chen, Y. Wang, and C.-C. J. Kuo, "Evaluating word embedding models: methods and experimental results," *APSIPA Transactions on Signal and Information Processing*, vol. 8, p. e19, 2019, doi: 10.1017/ATSIP.2019.12.

[12] D. Karani, "Introduction to word embedding and Word2Vec," *Towards Data Science*, vol. 1, 2018.

[13] Y. Li and T. Yang, "Word Embedding for Understanding Natural Language: A Survey," in *Guide to Big Data Applications*, S. Srinivasan, Ed., Cham: Springer International Publishing, 2018, pp. 83–104.

[14] K. Choudhary and R. Beniwal, "Xplore Word Embedding Using CBOW Model and Skip-Gram Model," in *2021 7th International Conference on Signal Processing and Communication (ICSC)*, 2021, pp. 267–270. doi: 10.1109/ICSC53193.2021.9673321.

[15] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient estimation of word representations in vector space," *arXiv preprint arXiv:1301.3781*, 2013.

[16] X. Rong, "word2vec Parameter Learning Explained." 2016.

[17] L. Ma and Y. Zhang, "Using Word2Vec to process big text data," in *2015 IEEE International Conference on Big Data (Big Data)*, 2015, pp. 2895–2897. doi: 10.1109/BigData.2015.7364114.

[18] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding." 2019.

[19] E. Fredkin, "Trie memory," *Commun. ACM*, vol. 3, no. 9, pp. 490–499, Sep. 1960, doi: 10.1145/367390.367400.

[20] X. Song, A. Salcianu, Y. Song, D. Dopson, and D. Zhou, "Fast WordPiece Tokenization," in *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, M.-F. Moens, X. Huang, L. Specia, and S. W.-t. Yih, Eds., Online and Punta Cana, Dominican Republic: Association for Computational Linguistics, Nov. 2021, pp. 2089–2103. doi: 10.18653/v1/2021.emnlp-main.160.

[21] C. Fellbaum and K. Brown, "Encyclopedia of language and linguistics." Elsevier Oxford, 2005.

[22] B. Liu, "Text sentiment analysis based on CBOW model and deep learning in big data environment," *Journal of ambient intelligence and humanized computing*, vol. 11, no. 2, pp. 451–458, 2020.

[23] E. S. Lubana, R. Dick, and H. Tanaka, "Beyond BatchNorm: Towards a Unified Understanding of Normalization in Deep Learning," in *Advances in Neural Information Processing Systems*, M. Ranzato, A. Beygelzimer, Y. Dauphin, P. Liang, and J. W. Vaughan, Eds., Curran Associates, Inc., 2021, pp. 4778–4791. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2021/file/2578eb9cdf020730f77793e8b58e165a-Paper.pdf

[24] L. Medsker and L. C. Jain, *Recurrent neural networks: design and applications*. CRC press, 1999.

[25] A. Graves, "Long Short-Term Memory," in *Supervised Sequence Labelling with Recurrent Neural Networks*, Berlin, Heidelberg: Springer Berlin Heidelberg, 2012, pp. 37–45. doi: 10.1007/978-3-642-24797-2_4.

[26] J. Chung, C. Gulcehre, K. Cho, and Y. Bengio, "Empirical Evaluation of Gated Recurrent Neural Networks on Sequence Modeling." 2014.

[27] A. Vaswani *et al.*, "Attention Is All You Need." 2023.

[28] OpenAI *et al.*, "GPT-4 Technical Report." 2024.

[29] S. L. Smith, A. Brock, L. Berrada, and S. De, "ConvNets Match Vision Transformers at Scale." 2023.

[30] R. Sutton, "The bitter lesson, 2019," *URL http://www. incompleteideas. net/IncIdeas/BitterLesson. html*, 2019.

[31] J. Kaplan *et al.*, "Scaling Laws for Neural Language Models." 2020.

[32] S. Deerwester, S. T. Dumais, G. W. Furnas, T. K. Landauer, and R. Harshman, "Indexing by latent semantic analysis," *Journal of the American Society for Information Science*, vol. 41, no. 6, pp. 391–407, 1990, doi: https://doi.org/10.1002/(SICI)1097-4571(199009)41:6<391::AID-ASI1>3.0.CO;2-9.

[33] V. Klema and A. Laub, "The singular value decomposition: Its computation and some applications," *IEEE Transactions on Automatic Control*, vol. 25, no. 2, pp. 164–176, 1980, doi: 10.1109/TAC.1980.1102314.

[34] A. Agrawal and A. An, "Unsupervised emotion detection from text using semantic and syntactic relations," in *2012 IEEE/WIC/ACM International Conferences on Web Intelligence and Intelligent Agent Technology*, 2012, pp. 346–353.

[35] S. D. Gollapalli, P. Rozenshtein, and S. K. Ng, "ESTeR: Combining word co-occurrences and word associations for unsupervised emotion detection," in *Findings of the Association for Computational Linguistics: EMNLP 2020*, 2020, pp. 1043–1056.

[36] Y.-c. Park, J.-G. Jang, and U. Kang, "Fast Partial Fourier Transform." 2020.

[37] L. Durak and O. Arikan, "Short-time Fourier transform: two fundamental properties and an optimal implementation," *IEEE Transactions on Signal Processing*, vol. 51, no. 5, pp. 1231–1242, 2003.

[38] L. Wang *et al.*, "Towards Learning Universal Audio Representations." 2022.

[39] Q. Kong, Y. Cao, T. Iqbal, Y. Wang, W. Wang, and M. D. Plumbley, "PANNs: Large-Scale Pretrained Audio Neural Networks for Audio Pattern Recognition." 2020.

[40] Y. M. Costa, L. S. Oliveira, and C. N. Silla, "An evaluation of Convolutional Neural Networks for music classification using spectrograms," *Applied Soft Computing*, vol. 52, pp. 28–38, 2017, doi: https://doi.org/10.1016/j.asoc.2016.12.024.

[41] M. French and R. Handy, "Spectrograms: Turning Signals into Pictures," *Journal of Engineering Technology*, vol. 24, pp. 32–35, 2007.

[42] P. Verma and J. Berger, "Audio Transformers:Transformer Architectures For Large Scale Audio Understanding. Adieu Convolutions." 2021.

[43] J. Wülfing and M. A. Riedmiller, "Unsupervised Learning of Local Features for Music Classification.," in *ISMIR*, 2012, pp. 139–144.

[44] Y.-C. Lu, C.-W. Wu, C.-T. Lu, and A. Lerch, "An Unsupervised Approach to Anomaly Detection in Music Datasets," in *Proceedings of the 39th International ACM SIGIR Conference on Research and Development in Information Retrieval*, in SIGIR '16. Pisa, Italy: Association for Computing Machinery, 2016, pp. 749–752. doi: 10.1145/2911451.2914700.

[45] M. Jogin, Mohana, M. S. Madhulika, G. D. Divya, R. K. Meghana, and S. Apoorva, "Feature Extraction using Convolution Neural Networks (CNN) and Deep Learning," in *2018 3rd IEEE International Conference on Recent Trends in Electronics, Information & Communication Technology (RTEICT)*, 2018, pp. 2319–2323. doi: 10.1109/RTEICT42901.2018.9012507.

[46] J. Selva, A. S. Johansen, S. Escalera, K. Nasrollahi, T. B. Moeslund, and A. Clapés, "Video Transformers: A Survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 11, pp. 12922–12943, 2023, doi: 10.1109/TPAMI.2023.3243465.

[47] F. Krause, *Video Emotion Classification utilizing Temporal Difference Images*. Master Thesis, Otto-von-Guericke University Magdeburg, 2023.

[48] A. Kumar, A. Kumar, V. Vineet, and Y. Rawat, "Benchmarking self-supervised video representation learning," p. , 2023.

[49] A. Arnab, M. Dehghani, G. Heigold, C. Sun, M. Lučić, and C. Schmid, "ViViT: A Video Vision Transformer." 2021.

[50] H. Akbari *et al.*, "VATT: Transformers for Multimodal Self-Supervised Learning from Raw Video, Audio and Text," *ArXiv*, 2021, [Online]. Available: https://api.semanticscholar.org/CorpusID:233346984

[51] G. Patil and P. Suja, "Emotion recognition from 3D videos using optical flow method," in *2017 International Conference On Smart Technologies For Smart Nation (SmartTechCon)*, 2017, pp. 825–829. doi: 10.1109/SmartTechCon.2017.8358488.

[52] J. Haddad, O. Lezoray, and P. Hamel, "3D-CNN for Facial Emotion Recognition in Videos," in *Advances in Visual Computing*, G. Bebis, Z. Yin, E. Kim, J. Bender, K. Subr, B. C. Kwon, J. Zhao, D. Kalkofen, and G. Baciu, Eds., Cham: Springer International Publishing, 2020, pp. 298–309.

[53] L. Zhang, Y. Tong, and Q. Ji, "Active Image Labeling and Its Application to Facial Action Labeling," in *Computer Vision – ECCV 2008*, D. Forsyth, P. Torr, and A. Zisserman, Eds., Berlin, Heidelberg: Springer Berlin Heidelberg, 2008, pp. 706–719.

[54] Y. Tong, W. Liao, and Q. Ji, "Facial Action Unit Recognition by Exploiting Their Dynamic and Semantic Relationships," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 29, no. 10, pp. 1683–1699, 2007, doi: 10.1109/TPAMI.2007.1094.

[55] B. Martinez, M. F. Valstar, B. Jiang, and M. Pantic, "Automatic Analysis of Facial Actions: A Survey," *IEEE Transactions on Affective Computing*, vol. 10, no. 3, pp. 325–347, 2019, doi: 10.1109/TAFFC.2017.2731763.

[56] M. Valstar, M. Pantic, and I. Patras, "Motion history for facial action detection in video," in *2004 IEEE International Conference on Systems, Man and Cybernetics (IEEE Cat. No.04CH37583)*, 2004, pp. 635–640. doi: 10.1109/ICSMC.2004.1398371.

[57] H. Shimodaira, K.-i. Noma, M. Nakai, and S. Sagayama, "Dynamic Time-Alignment Kernel in Support Vector Machine," in *Advances in Neural Information Processing Systems*, T. Dietterich, S. Becker, and Z. Ghahramani, Eds., MIT Press, 2001, p. . [On-

line]. Available: https://proceedings.neurips.cc/paper_files/paper/2001/file/a869ccbcbd9568808b8497e28275c7c8-Paper.pdf

[58] F. Zhou, F. De la Torre, and J. F. Cohn, "Unsupervised discovery of facial events," in *2010 IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, 2010, pp. 2574–2581. doi: 10.1109/CVPR.2010.5539966.

[59] T. Warren Liao, "Clustering of time series data—a survey," *Pattern Recognition*, vol. 38, no. 11, pp. 1857–1874, 2005, doi: https://doi.org/10.1016/j.patcog.2005.01.025.

[60] S. Poria, D. Hazarika, N. Majumder, G. Naik, E. Cambria, and R. Mihalcea, "Meld: A multimodal multi-party dataset for emotion recognition in conversations," *arXiv preprint arXiv:1810.02508*, 2018.

[61] C. Busso *et al.*, "IEMOCAP: Interactive emotional dyadic motion capture database," *Language resources and evaluation*, vol. 42, pp. 335–359, 2008.

[62] V. Vielzeuf, S. Pateux, and F. Jurie, "Temporal multimodal fusion for video emotion classification in the wild," in *Proceedings of the 19th ACM International Conference on Multimodal Interaction*, in ICMI '17. Glasgow, UK: Association for Computing Machinery, 2017, pp. 569–576. doi: 10.1145/3136755.3143011.

[63] A. Dhall, R. Goecke, S. Lucey, and T. Gedeon, "Collecting Large, Richly Annotated Facial-Expression Databases from Movies," *IEEE MultiMedia*, vol. 19, no. 3, pp. 34–41, 2012, doi: 10.1109/MMUL.2012.26.

[64] A. S. Patwardhan, "Multimodal mixed emotion detection," in *2017 2nd International Conference on Communication and Electronics Systems (ICCES)*, 2017, pp. 139–143. doi: 10.1109/CESYS.2017.8321250.

[65] M. G. Huddar, S. S. Sannakki, and V. S. Rajpurohit, "Multi-level feature optimization and multimodal contextual fusion for sentiment analysis and emotion classification," *Computational Intelligence*, vol. 36, no. 2, pp. 861–881, 2020, doi: https://doi.org/10.1111/coin.12274.

[66] C. Yang, F. Chen, and S. Tian, "Grouped Genetic Algorithm Based Optimal Tests Selection for System with Multiple Operation Modes," *Journal of Electronic Testing*, vol. 33, p. , 2017, doi: 10.1007/s10836-017-5672-y.

[67] N. Majumder, D. Hazarika, A. Gelbukh, E. Cambria, and S. Poria, "Multimodal sentiment analysis using hierarchical fusion with context modeling," *Knowledge-Based Systems*, vol. 161, pp. 124–133, 2018, doi: https://doi.org/10.1016/j.knosys.2018.07.041.

[68] H. Pham, T. Manzini, P. P. Liang, and B. Poczós, "Seq2Seq2Sentiment: Multimodal Sequence to Sequence Models for Sentiment Analysis," in *Proceedings of Grand Challenge and Workshop on Human Multimodal Language (Challenge-HML)*, A. Zadeh, P. P. Liang, L.-P. Morency, S. Poria, E. Cambria, and S. Scherer, Eds., Melbourne, Australia: Association for Computational Linguistics, Jul. 2018, pp. 53–63. doi: 10.18653/v1/W18-3308.

[69] J. Pennington, R. Socher, and C. D. Manning, "Glove: Global vectors for word representation," in *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)*, 2014, pp. 1532–1543.

[70] G. Degottex, J. Kane, T. Drugman, T. Raitio, and S. Scherer, "COVAREP — A collaborative voice analysis repository for speech technologies," in *2014 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2014, pp. 960–964. doi: 10.1109/ICASSP.2014.6853739.

[71] A. Zadeh, M. Chen, S. Poria, E. Cambria, and L.-P. Morency, "Tensor Fusion Network for Multimodal Sentiment Analysis." 2017.

Appendix E.

Appendix

Statement of Authorship

I herewith assure that I wrote the present thesis independently, that the thesis has not been partially or fully submitted as graded academic work and that I have used no other means than the ones indicated. I have indicated all parts of the work in which sources are used according to their wording or to their meaning.

I am aware of the fact that violations of copyright can lead to injunctive relief and claims for damages of the author as well as a penalty by the law enforcement agency.

Magdeburg, September 13, 2024

_____

Signature